Hand Gesture Controlled Robot using TinyML

INSTITUTE OF SCIENCE & TECHNOLOGY
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Problem Statement

Controlling robots using traditional methods such as joysticks or remote controllers can be hard-to-use and restrictive. These systems often require physical interfaces, which limit ease of use and natural interaction. There is a growing need for a more intuitive and contactless method of controlling robotic systems, especially in applications like assistive technology and smart automation.

Objective

1. To design a robot that can be controlled using simple hand gestures
2. To implement real-time gesture recognition using TinyML on an embedded system
3. To enable reliable wireless communication between transmitter and receiver
4. To develop a compact, low-cost, and efficient robotic system
5. To provide smooth and responsive motion based on user input

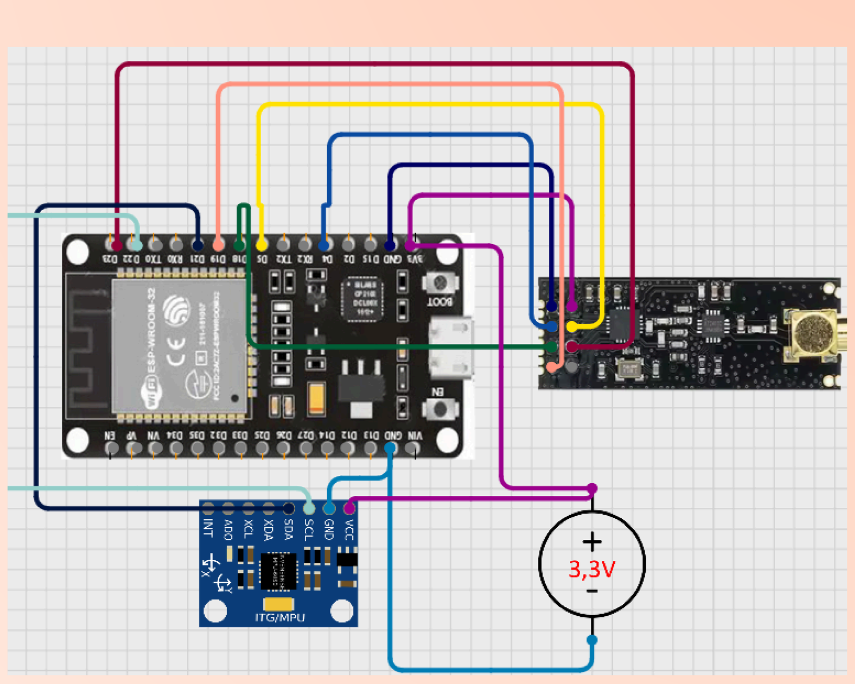
Methodology and Design

The system is divided into two main sections: transmitter and receiver.

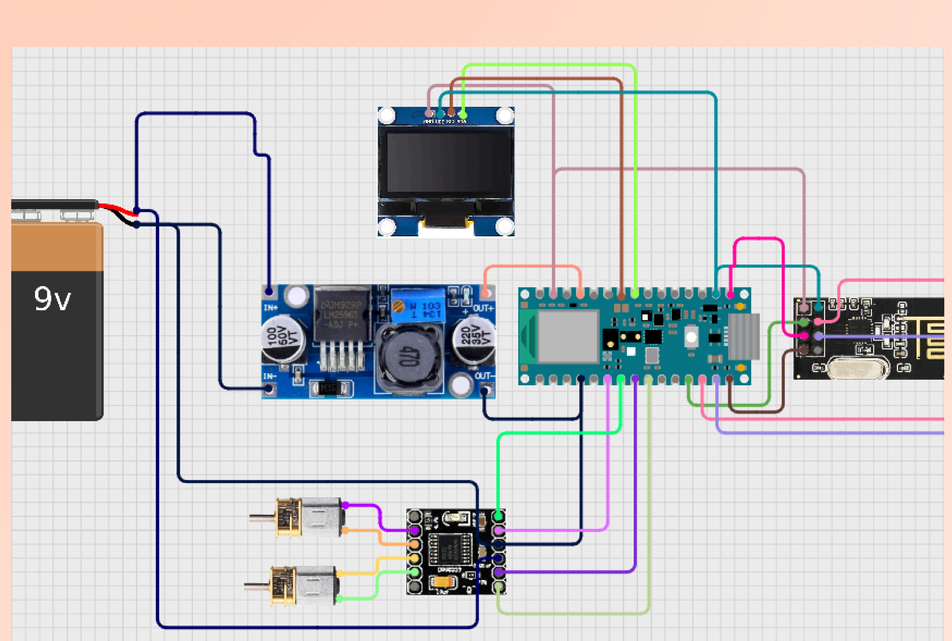
- On the transmitter side, hand movements are captured using an MPU6050 sensor, which measures acceleration and orientation. This data is processed by an ESP32 microcontroller, where a trained TinyML model (developed using Edge Impulse) classifies the gestures.

- Once a gesture is recognized, the corresponding command is transmitted wirelessly using an NRF24L01 module.

- On the receiver side, an Arduino Nano receives the transmitted data. Based on the received command, the DRV8833 motor driver controls two N20 DC motors to move the robot in the desired direction. An OLED display is used to show the current movement direction for feedback.



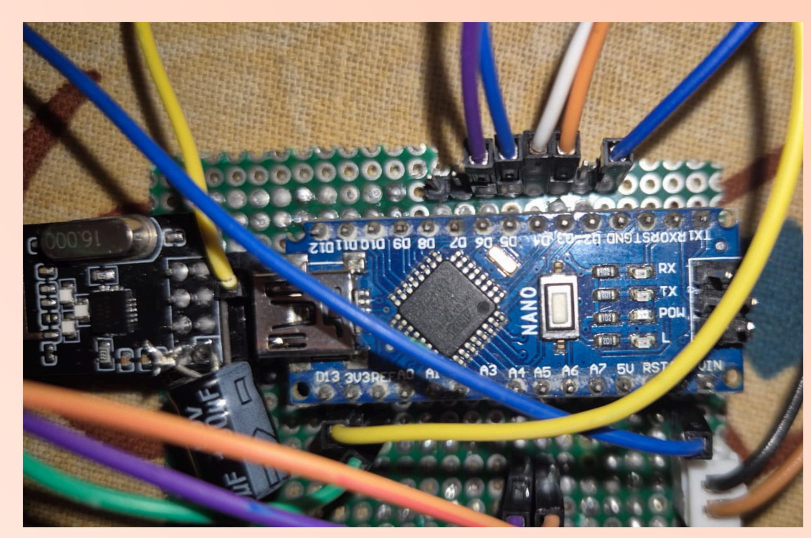
Transmitter Side Circuit

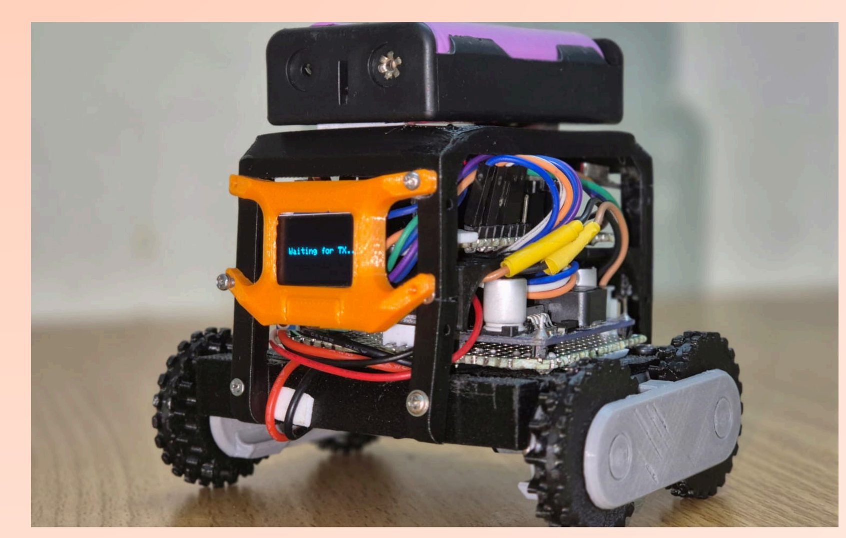


Reciever Side Circuit

Hardware Overview

- ESP32 Microcontroller : Acts as the core processing unit of the system. It reads sensor data, runs the TinyML model for gesture classification, and manages wireless communication. Its higher processing capability and built-in support for embedded AI make it suitable for real-time inference.
- Arduino Nano : Serves as the control unit on the receiver side. It processes incoming data from the NRF24L01 and generates control signals for motor operation. Its compact size and simplicity make it ideal for embedded applications.
- MPU6050 (Accelerometer + Gyroscope) : A 6-axis motion tracking sensor used to capture hand gestures. It provides both linear acceleration and angular velocity data, enabling accurate detection of movement and orientation.
- NRF24L01 Transceiver Module : Handles wireless communication between the transmitter and receiver. It operates on the 2.4 GHz band and enables low-power, low-latency data transmission for real-time control.
- DRV8833 Motor Driver : A dual H-bridge motor driver used to control the direction and speed of two DC motors. It provides efficient current handling and supports bidirectional motor control.





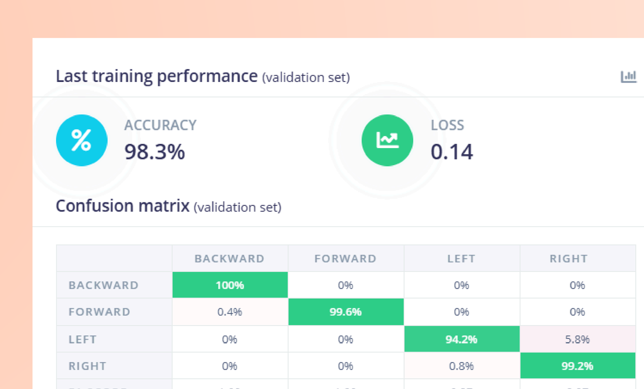
Results and Output

The model was trained using real-time motion data from the MPU6050 and achieved strong performance on the validation set. The system reached an accuracy of 98.3% with low loss (0.14), indicating reliable gesture classification.
The confusion matrix shows that most gestures are correctly identified, with very minimal misclassification between similar movements. The F1 scores (~0.97–1.00) further confirm balanced and consistent model performance across all classes.
The data explorer plot shows clear clustering of gesture data, indicating that the model is able to effectively separate different motion patterns. Only a few overlapping points are observed, which explains the small classification errors.
From the feature extraction stage, the filtered signals and spectral analysis (FFT) highlight distinct patterns for each gesture, helping the model learn meaningful differences in motion.
On deployment, the model runs efficiently on the ESP32 with:

- Inference time: ~1 ms
- Low memory usage (RAM & Flash)

In real-time operation:

- Gestures are accurately recognized
- Commands are transmitted wirelessly with minimal delay
- The robot responds smoothly in all directions
- OLED display correctly shows the detected movement



Last training performance (validation set)

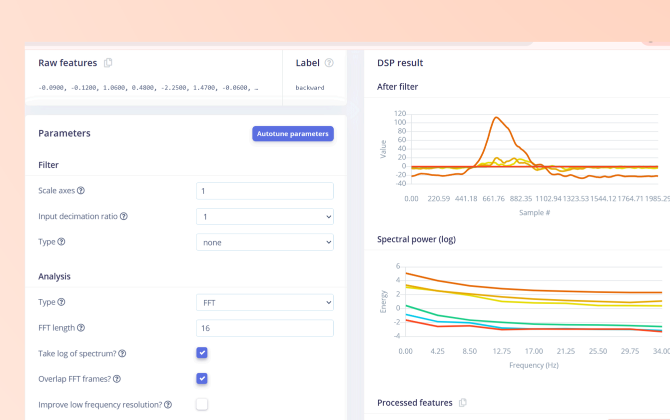
ACCURACY 98.3%

LOSS 0.14

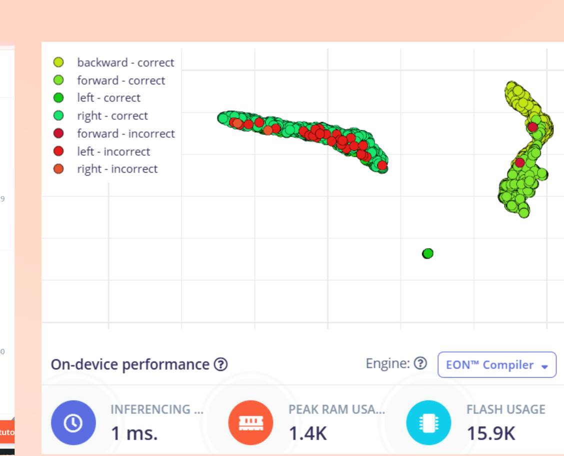
Confusion matrix (validation set)

	BACKWARD	FORWARD	LEFT	RIGHT
BACKWARD	100%	0%	0%	0%
FORWARD	0.4%	99.6%	0%	0%
LEFT	0%	0%	94.2%	5.8%
RIGHT	0%	0%	0.8%	99.2%
F1 SCORE	1.00	1.00	0.97	0.97

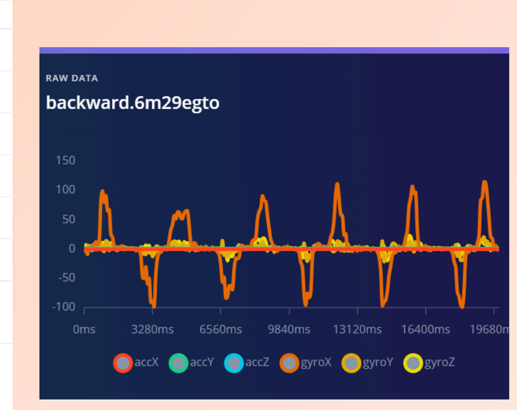
Confusion Metrics



Features Extraction and Signal processing



Data Explorer Plot



Data Acquisitions

Conclusion

This project presents a gesture-controlled robot using TinyML for real-time, on-device decision making. The system accurately converts hand gestures into robot movements with minimal delay and reliable wireless communication.By eliminating the need for traditional controllers and cloud processing, it offers a faster, low-power, and user-friendly solution.